Experiment 1

AIM: Explore GitHub/GitLab for open-source projects with different licenses

1. **List of GitHub Licenses**

License	What it is	You can…	You must…	Best for
No License	Default (no permissions granted)	View the code only	Ask permission to use, copy, or modify	Private repos, code you don't want reused
MIT License	Very permissive	Use, copy, modify, sell, sublicense	Keep copyright + license notice	Open-source libraries, maximum adoption
Apache License 2.0	Permissive + patent protection	Use commercially, modify, distribute	Include license + notice, state changes	Corporate-friendly open source
BSD 2-Clause	Minimal permissive	Almost anything	Keep copyright notice	Simple libraries, academic code
BSD 3-Clause	Permissive (with name protection)	Same as BSD 2-Clause	Don't use author name for promotion	Safer branding protection
Boost Software License 1.0	Permissive, simple	Use in any project	Include license text	C++ libraries, header-only code
GNU GPL v3.0	Strong copyleft	Use, modify, distribute	Open-source your changes under GPL	Software meant to stay open
GNU AGPL v3.0	Network copyleft	Same as GPL	Release source even for SaaS	Web apps & services
Eclipse Public License 2.0	Weak copyleft	Use commercially	Open-source modifications only	Enterprise & tooling ecosystems
Creative Commons Zero (CC0)	Public domain	Do absolutely anything	Nothing	Snippets, datasets, examples

2. Licenses and Short Description Table

License Name	Description
No License	Code is private; others can only view it and need permission to use or modify
MIT License	Very permissive license allowing reuse with minimal restrictions
Apache License 2.0	Permissive license with added patent protection
BSD 2-Clause	Simple permissive license with minimal requirements
BSD 3-Clause	Permissive license that prevents author name endorsement
Boost Software License 1.0	Business-friendly license, common for C++ projects
GNU GPL v3.0	Strong copyleft license requiring all derivatives to remain open-source
GNU AGPL v3.0	Copyleft license that applies even to web and SaaS applications
Eclipse Public License 2.0	Weak copyleft license; only modified files must be open-sourced
CC0 1.0	Public domain license with no restrictions or attribution required

3. Licenses Comparison Table

License	Type	Commercial Use	Modify	Distribute	Must Open-Source Changes?	Patent Protection	Special Notes
No License	None	❌	❌	❌	❌	❌	Others can only *view* the code
MIT License	Permissive	✅	✅	✅	❌	❌	Very simple & popular
Apache License 2.0	Permissive	✅	✅	✅	❌	✅	Strong patent protection
BSD 2-Clause	Permissive	✅	✅	✅	❌	❌	Minimal restrictions
BSD 3-Clause	Permissive	✅	✅	✅	❌	❌	Prevents name endorsement
Boost Software License 1.0	Permissive	✅	✅	✅	❌	❌	Extremely business-friendly
GPL v3.0	Strong Copyleft	✅	✅	✅	✅ (all derivatives)	✅	Keeps software open forever
AGPL v3.0	Strong Copyleft (Network)	✅	✅	✅	✅ (even SaaS)	✅	Covers web services
Eclipse Public License 2.0	Weak Copyleft	✅	✅	✅	⚠️ (only modified files)	✅	Enterprise-friendly copyleft
CC0 1.0	Public Domain	✅	✅	✅	❌	❌	No attribution required

4. List of GitHub Alternatives

Platform Name	Description
GitLab	Open-source DevOps platform with built-in CI/CD
Bitbucket	Git repository hosting by Atlassian, good for teams
SourceForge	Hosting for open-source software projects
Gitea	Lightweight, self-hosted Git service
Gogs	Simple and fast self-hosted Git platform
Azure DevOps	Microsoft's platform for code, CI/CD, and planning
AWS CodeCommit	Secure Git hosting service by Amazon
Launchpad	Code hosting platform mainly for Ubuntu projects
Phabricator	Developer collaboration platform with code review
Codeberg	Community-driven, open-source Git hosting

5. **GitHub vs GitLab Table**

Feature	GitHub	GitLab
Ownership	Microsoft	GitLab Inc.
Hosting Type	Cloud-based	Cloud & self-hosted
Open Source	Partially open-source	Fully open-source (core)
CI/CD	Uses GitHub Actions	Built-in CI/CD
Repository Management	Excellent for open source	Strong for DevOps workflows
Issue Tracking	Basic	Advanced
Security Features	Good	Strong (DevSecOps)
Community	Very large	Growing
Best For	Open-source collaboration	Complete DevOps lifecycle

6. **Open-Source vs Proprietary vs Freeware**

Feature	Open-Source Software	Proprietary Software	Freeware
Source Code	Available to everyone	Not available	Not available
Cost	Free or paid	Usually paid	Free
Modify Code	✅ Allowed	❌ Not allowed	❌ Not allowed
Redistribution	✅ Allowed	❌ Restricted	❌ Restricted
Ownership	Community or organization	Single company	Single company
Transparency	High	Low	Low
Examples	Linux, Git, Mozilla Firefox	Windows, MS Office, Photoshop	Google Chrome, Skype